

Investigation of Multi-Variate Datasets

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1 Glass Identification

The Glass Identification dataset (B. German, 1987) consists of nine continuous variables, one categorical variable (type of glass) and 214 instances. The nine continuous variables (features) are the glass's refractive index (RI) and eight different oxides measured in weight percent. Motivated by criminal investigations, this dataset aims to enable the identification of types of glass.

This investigation seeks to determine, by visual assessment, two oxides in glass that best predict the RI and two oxides that best predict the type of glass. In doing so, it will inform which two oxides should be prioritised for measurement to correctly identify the type of glass.

A notable aspect of the dataset is its complete absence of instances of *Type 4* glass, as shown in [Figure 1](#). Therefore any conclusions on glass type prediction must be caveated by the unavailability of data on how the oxides vary in *Type 4* glass. In addition, the seven types of glass, although represented by integers (1 to 7), are unordered factor levels and will be treated as such in the following analysis.

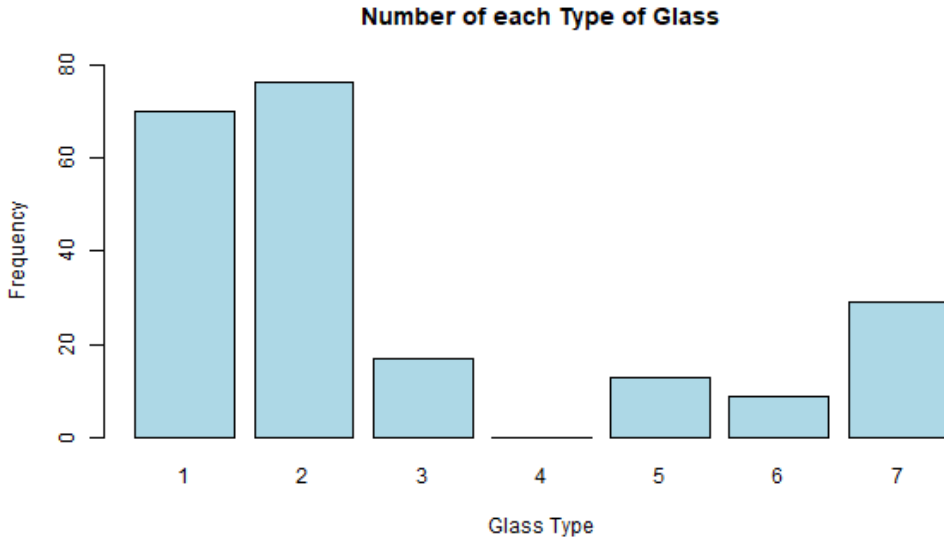


Figure 1: A bar chart of the number of each type of glass in the Glass Identification dataset.

The two oxides in the glass that best predict the RI were determined by analysis of variance (ANOVA) from the built-in R function `aov()` (Venables and Smith, 2024, p. 54). [Code Listing 1](#) shows how this function was implemented and its results plotted to produce the heatmap in [Figure 2](#). The heatmap shows that the greatest R^2 value occurs at the intersection of Si and Ca, indicating that their corresponding oxides explain the greatest proportion of variation RI. This pairing was further validated as the best predictor by inspecting the p-values associated with all possible pairwise interactions in an ANOVA. The Si:Ca interaction has a p-value of 0.00398 which is highly significant at $\alpha = 0.05$.

[Figure 3](#) is a multivariate plot showing how the variation of RI with Si depends on the value Ca. The bottom right of the plot has lower values of RI (dark blue) which gradually increase (to yellow) as one moves to the top left of the plot. This is indicative of the interaction between Si and Ca in the prediction of RI.

The two oxides that best predict the type of glass were determined by fitting a multinomial logistic regression model. This was necessary for this task as the type of glass is a categorical variable with more than two unordered factors. [Figure 4](#) shows that the pairing with the lowest Akaike Information Criterion (AIC) is Mg and Ca. The model for predicting the type of glass using just Mg and Ca is therefore most likely to be correct and we

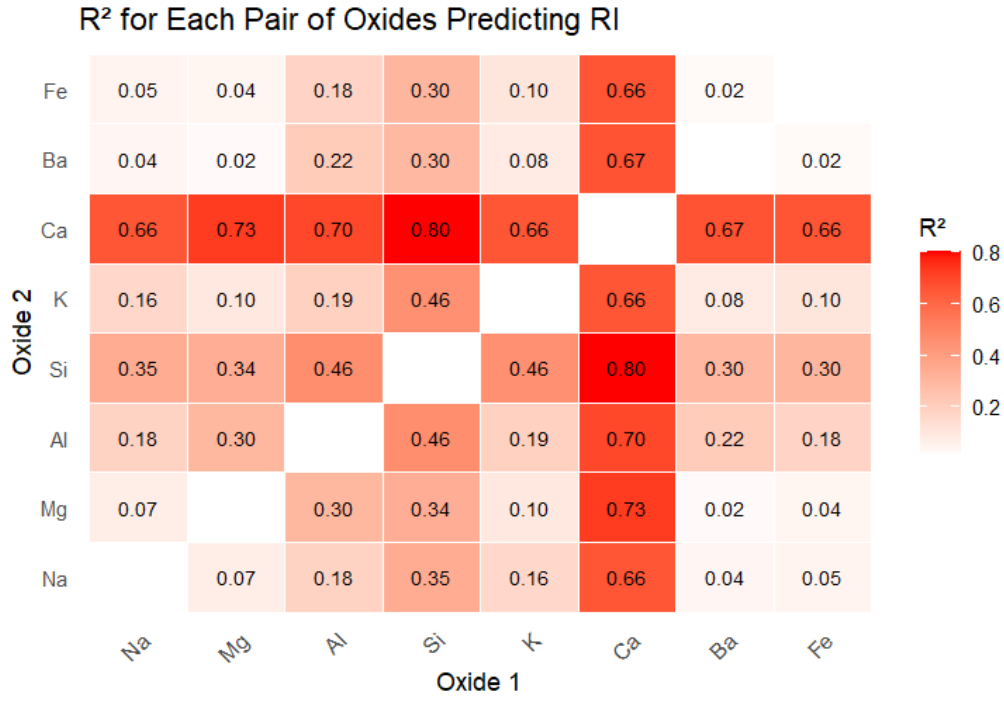


Figure 2: A heatmap of R^2 values for all possible pairings of oxides predicting refractive index.

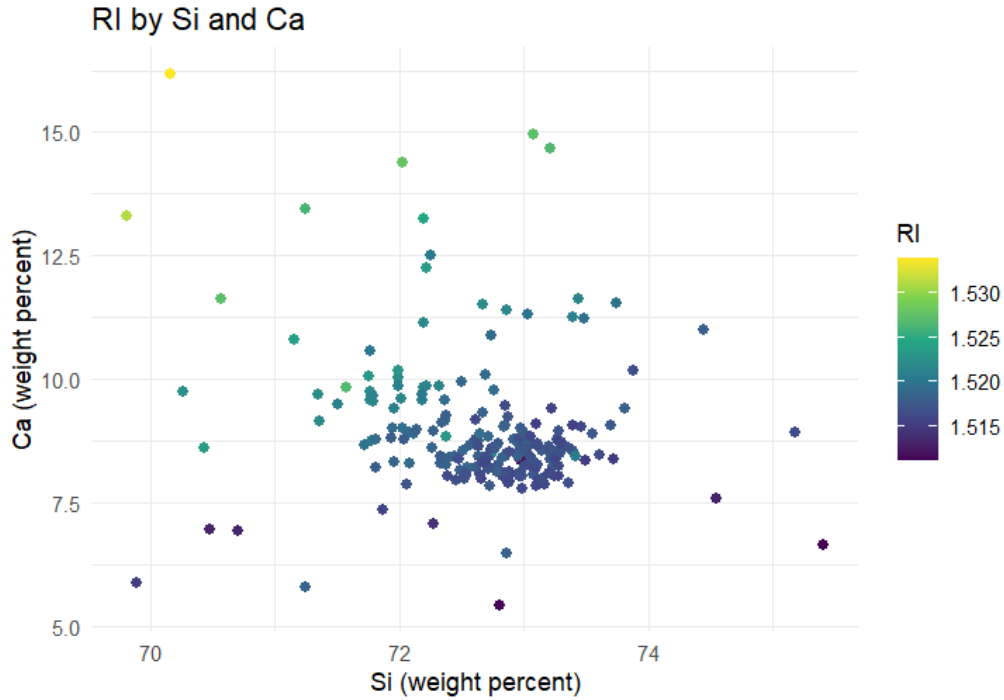


Figure 3: A multivariate plot showing how Si (weight percent) and Ca (weight percent) vary in relation to the refractive index which is denoted by colour.

conclude that Mg and Ca are the best oxides in predicting the type of glass. However, it should be noted that the Mg and Al pairing has an AIC value of just 1.4 lower than that of Mg and Ca. Models that have similar AIC values (difference less than 1) cannot be distinguished so we conclude that Mg and Ca are indeed the best pairing to predict the type of glass - although Mg and Al are a close second (Perley, 2024, p. 51).

The weight percent of Mg and Ca vary greatly across the different types of glass as can be seen in [Figure 5](#). *Type 5* glass has the highest median for Ca and *Type 2* glass has the highest median for Mg. Overall the variation in both Mg and Ca between the types of glass highlights their importance in predictions. In particular, Mg is good at distinguishing *Type 6* glass and Ca *Type 5* glass.

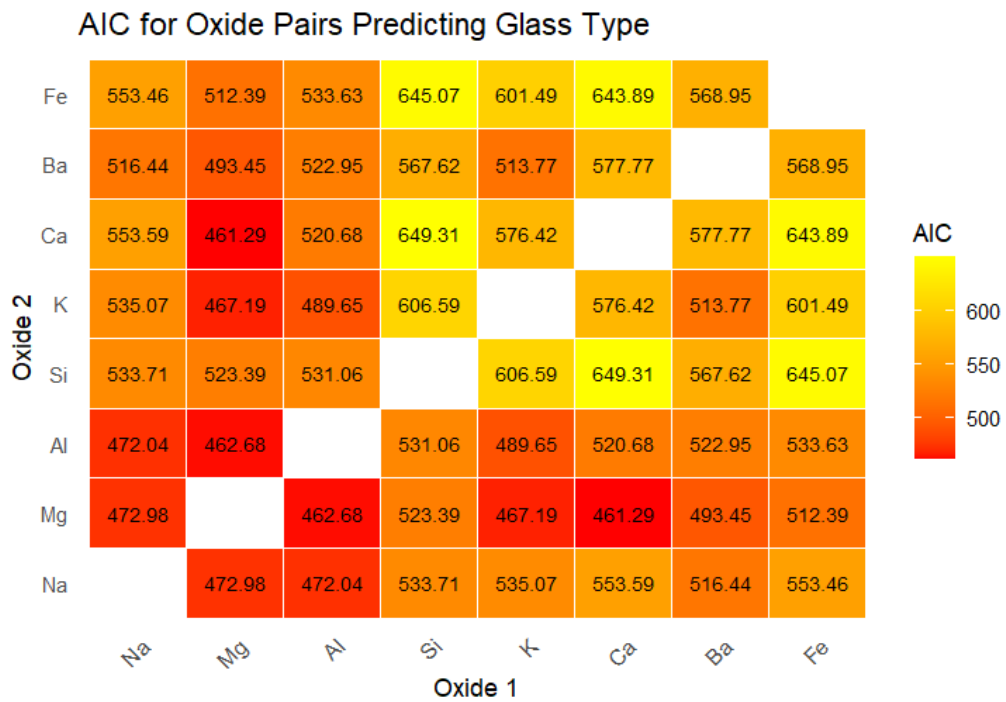


Figure 4: A heatmap of Akaike Information Criterion (AIC) values for all possible pairings of oxides predicting type of glass.

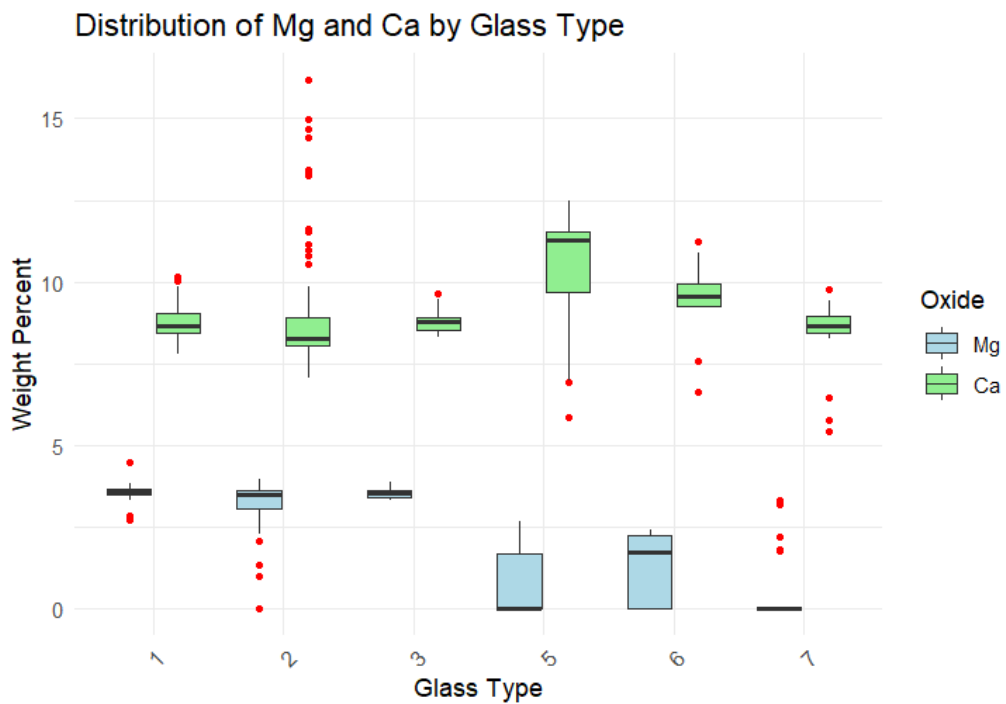


Figure 5: A multivariate plot showing how Mg (weight percent) and Ca (weight percent) vary in relation to the type of glass which is denoted by colour.

2 El Nino

The El Nino dataset contains oceanographic and surface meteorological measurements taken from 70 buoys in the equatorial Pacific Ocean from 7th March 1980 to 23rd June 1998. Its purpose is to aid the detection, understanding and prediction of climate variations relating to the El Nino/Southern Oscillation (ENSO) cycles (Cook, 1999). 11 variables make up the dataset - seven continuous variables, three integer variables, and one date variable. These variables are latitude, longitude, day, month, year, date, zonal winds, meridional winds, humidity, air temperature and sea surface temperature. This analysis will explore the key features of the air and sea surface temperature variations and how these variations depend on position.

Notably, both the air and sea surface temperatures fluctuated between 20 and 30 degrees Celcius, and both the zonal and meridional winds fluctuated between $-10m/s$ and $+10m/s$. Furthermore, the *Additional Variable Information* section in the dataset (Cook, 1999) states that there is no linear relationship between any of the wind variables and the other three meteorological variables (humidity, air temperature and sea surface temperature). Therefore, these relationships will not be explored in the following analysis.

Both air and sea surface temperatures are in the same range as mentioned earlier, but sea surface temperature skews towards the higher end compared to air temperature (see Figure 6). The most notable that can be seen in Figure 6 is the significant overlap between the two variables. This overlap suggests a relationship between air and sea surface temperature.

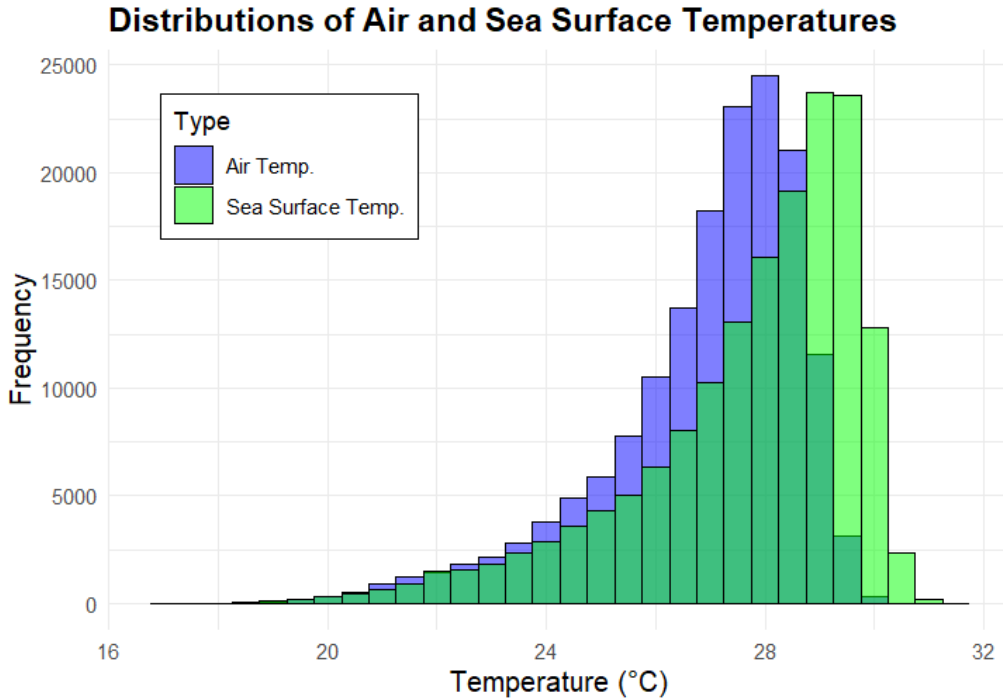


Figure 6: Overlapping histograms for air temperature and sea surface temperature from the El Nino dataset. Air temperature is shown in blue and sea surface temperature in green.

This relationship was explored further by carrying out a correlation test. The Pearson correlation coefficient was determined to be 0.94 with a p-value of $< 2.2 \times 10^{-16}$, which is highly statistically significant at $\alpha = 0.05$. Therefore, we can conclude that there is a strong, positive linear relationship between the air temperature and sea surface temperature, as depicted in Figure 7.

The air and sea surface temperatures both vary with position as is seen in Figure 8 and Figure 9. First, note that the grid-like pattern likely stems from the buoy placement systematically covering particular longitudinal and latitudinal positions. This is simply a feature of the measurement method rather than a feature of the climate. We expect both plots to be similar as we already concluded that air temperature and sea surface temperature are highly correlated. Both graphs also show that higher temperatures tend to be clustered near the equator and temperatures decrease as distance increases from the equator. This is expected as the equator receives more sunlight per unit area than areas away from the equator. Also, note that higher temperatures tend to be at longitude values of less than around 200.

To measure the air and sea surface temperatures' spatial dependence, ANOVA were performed to characterise the extent to which air and sea surface temperatures vary with position. It was determined that the ANOVA carried out on air temperature returned estimates and p-values of 2.33×10^{-3} and 0.00171 for latitude, and -3.264×10^{-2} and $< 2 \times 10^{-16}$ for longitude, respectively. These are both statistically significant at $\alpha = 0.05$.

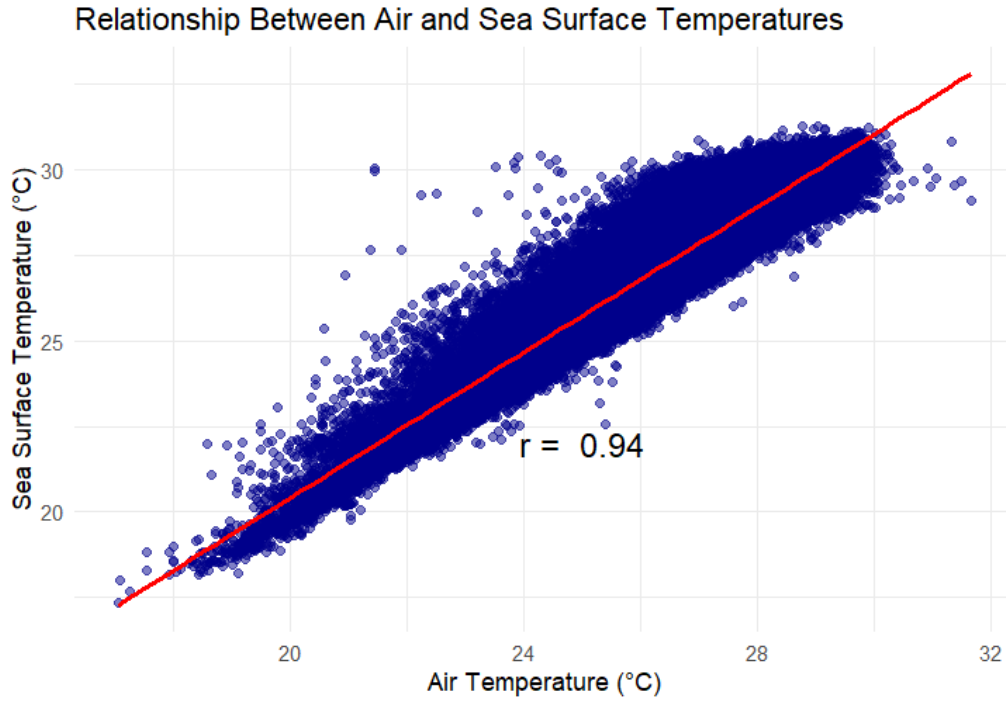


Figure 7: Scatter plot of sea surface temperature against air temperature with a solid red line for the linear model fit. The Pearson correlation coefficient, r , is overlaid on the plot.

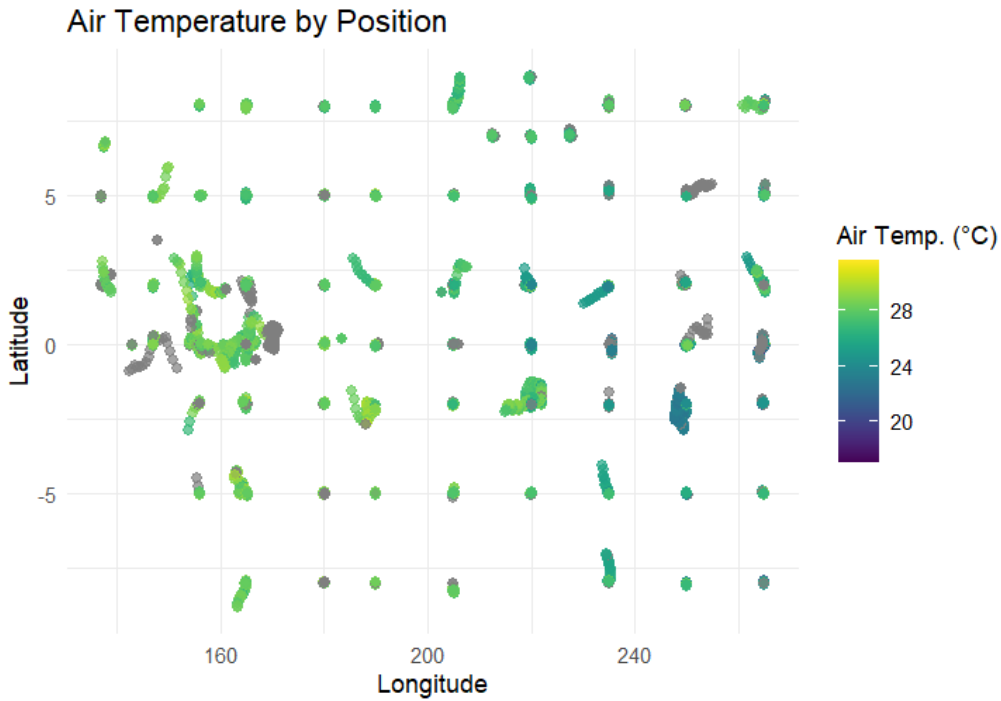


Figure 8: Scatter plot of sea surface temperature against air temperature with a red line for the linear model fit. The Pearson correlation coefficient, r , is overlaid on the plot.

and we can conclude that air temperature increases with latitude and decreases with longitude. Similarly for sea surface temperature, the ANOVA returned estimates and p-values of 0.0232 and $< 2 \times 10^{-16}$ for latitude, and -0.0375 and $< 2 \times 10^{-16}$ for longitude, respectively. Furthermore, the R^2 values for each were 0.4295 and 0.4614, respectively, meaning that position accounts for 43% and 46% of the variation in air temperature and sea surface temperature, respectively. Understanding of how meteorological variables vary with position is crucial for the avoidance of natural disasters that have been known to arise from ENSO cycles.

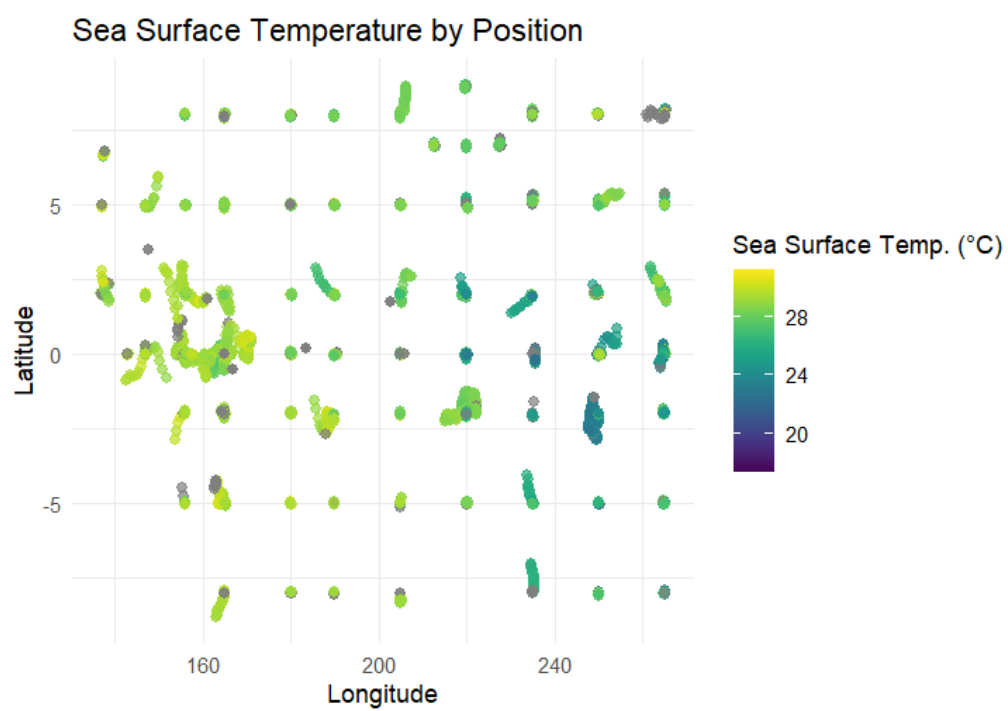


Figure 9: Scatter plot of sea surface temperature against air temperature with a red line for the linear model fit. The Pearson correlation coefficient, r , is overlaid on the plot.

3 Adult

The Adult dataset is also known as “Census Income” dataset and is used to predict whether the annual income of an adult exceeds \$50K/yr based on 14 features gathered from census data (Barry Becker, 1996). This dataset was chosen as it has real-world applications in classification tasks and has a good mix of both categorical and continuous variables.

The target variable (or class) is binary: either annual salary is $\leq$ \$50K or $>$ \$50K. Another interesting aspect of this dataset is that education is encoded in two ways. The first is education as a factor with levels such as “Bachelors” and “HS-grad”. The second is an integer variable called education-num which encodes the different education levels as integers with a meaningful order, e.g. “Doctorate” is greater than “Bachelors” which is greater than “Preschool”.

One of the categorical variables is called *workclass* and describes the type of job that the adult does. Two factor levels in this variable, “never-worked” and “without-pay” caused near-perfect separation when fitting a logistic regression model to the data. This is as expected since an adult who has never worked or is without pay is extremely likely to have an income below the threshold. To avoid separation, these two factor levels were removed from the dataset as they were obscuring other relationships in the data.

Another problem encountered was overfitting due to the low cell counts in some factor levels of the *native-country* variable. Factor levels in *native-country* that had less than 30 instances were removed from the dataset to prevent overfitting.

Figure 10 shows that many people with a low education earn an annual salary below the threshold. It also shows that few people in the higher range of education earn below the threshold.

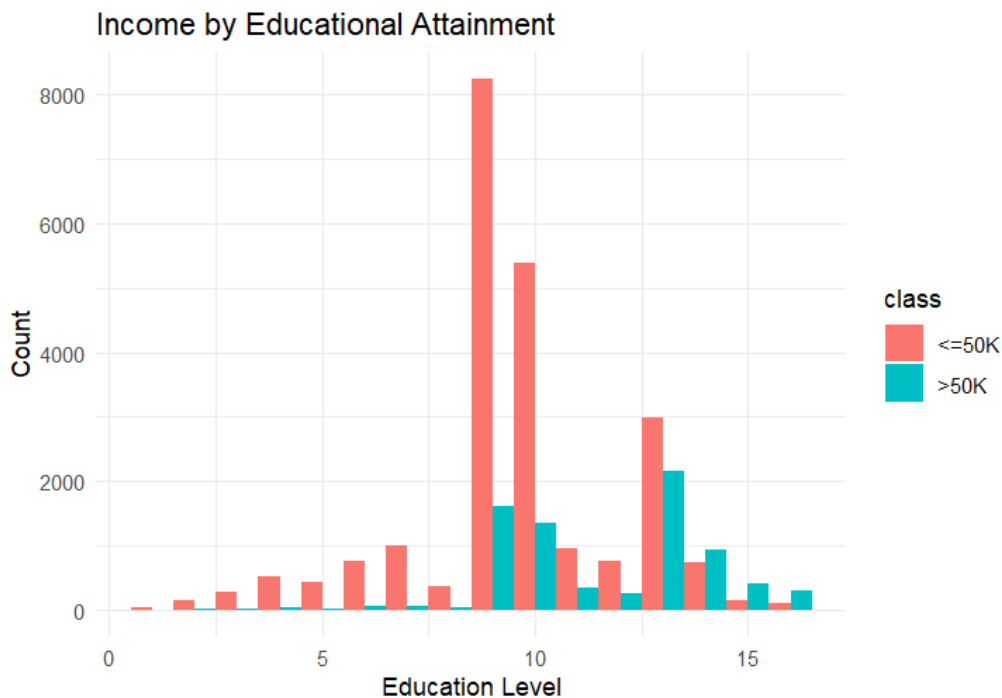


Figure 10: Histogram of education level organised by above or below salary threshold of \$50K.

Interestingly, Figure 11 shows that both groups have their modal working hours per week at around 40. This suggests that the number of hours an adult in the US works per week is not a good predictor of which side of the salary threshold they fall. Figure 12 reinforces this idea, but gives further insight. It shows that capital gain is a good predictor of the target variable, especially at values for capital gain above 50,000 as there is not a single individual below the salary threshold above that capital gain value. Furthermore, even for moderate capital gain between 10,000 and 50,000, capital gain is still a good predictor for the target variable.

Figure 13 shows how the target variable depends on the combination of education and hours worked per week. It shows that adults with the lowest and highest education levels work the most and that there is no interaction between hours worked and education in predicting the target variable.

This dataset gives great insights into the American population’s relationship with income. For this dataset to improve, a true copy of the census to avoid the overfitting issues discussed earlier would be useful.

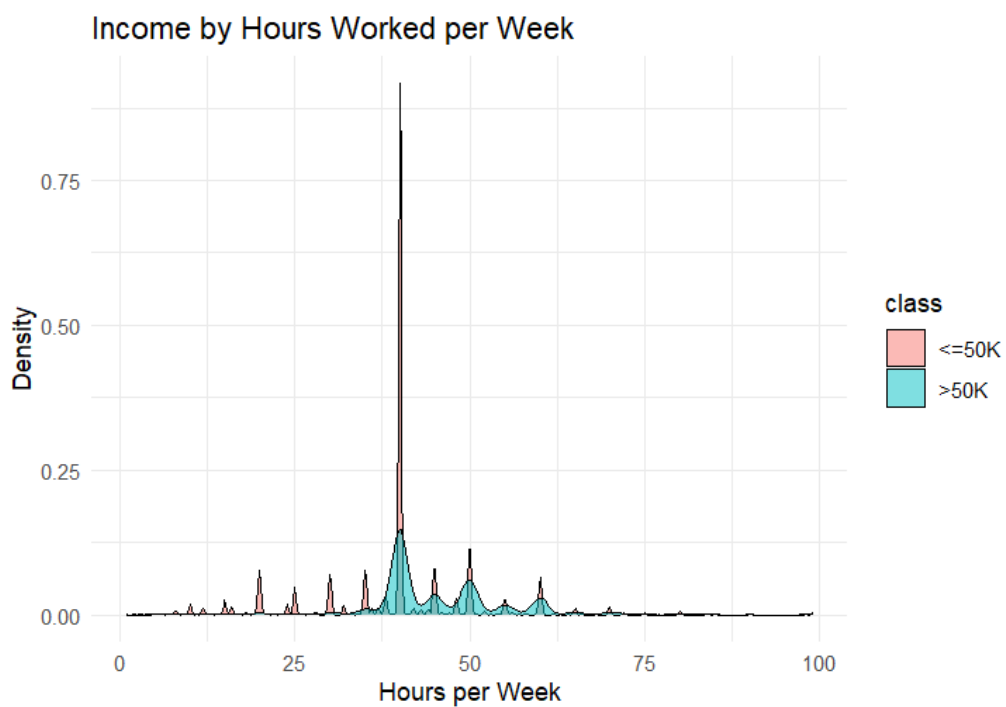


Figure 11: Density plot of hours per week worked organised by above or below salary threshold of \$50K.

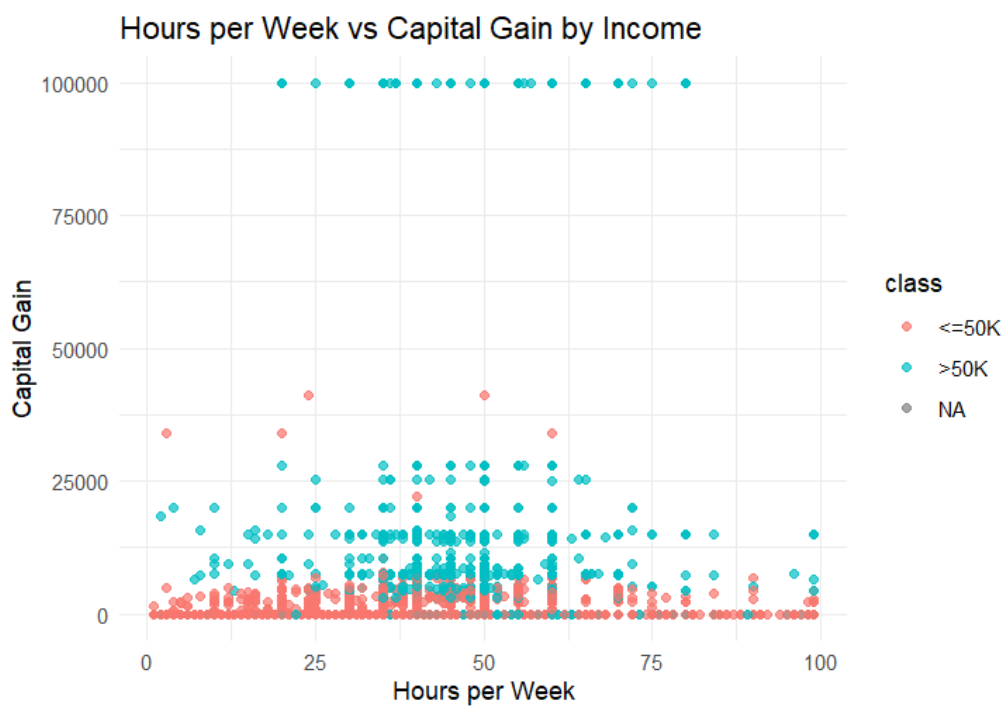


Figure 12: Multivariate scatter plot of hours per week worked against capital gain organised by above or below salary threshold of \$50K.

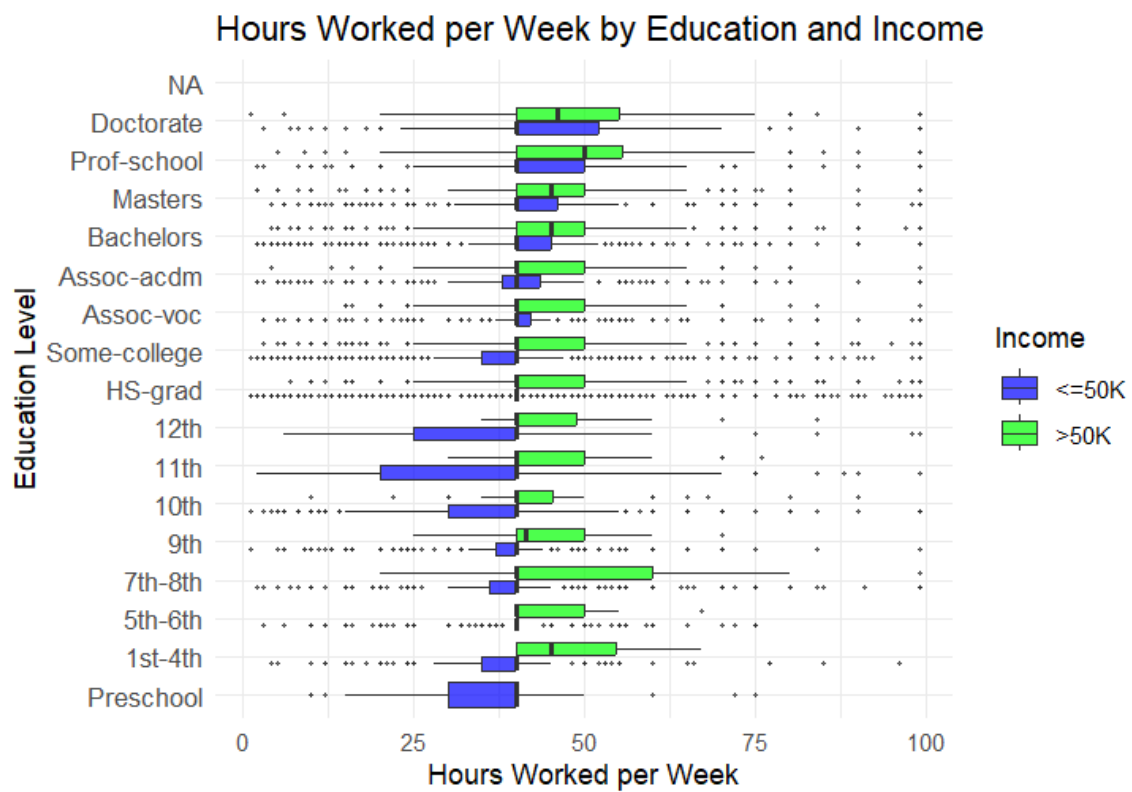


Figure 13: Multivariate boxplot showing how income class varies with education and hours worked per week.

4 Dow Jones Index

- Description of data, including why you chose the dataset, and any notable aspects. [4] The Dow Jones Index dataset contains historical financial data to predict the change in a stock's price for the next week since each row in the dataset is a week (Brown, 2013). The dataset has 15 features and 750 instances so it meets the criteria for the assignment. Furthermore, it covers many companies so can give a good induction of the state of many industries. An interesting feature in this dataset is that on 5/27/2011, every stock in the Dow Jones Index recorded a loss which highlights periods of market anomalies.

Here we will focus on a binary classification model by adding a column to the dataset that is simply 1 if it predicts the stock's price will increase next week and 0 otherwise. This is still practically sound as some applications only need to make money and are not worried about the amount.

Figure 14 displays the average percent change in price for the current week (blue) and the following week (red) over time. As expected, the red line lags behind in its shape by about a week compared to the blue line. An important feature to notice is towards the end of June the red line spikes significantly, signalling market volatility. The lag between the lines allows for predictive insights and may be able to help understand the behaviour of stock prices in the following weeks.

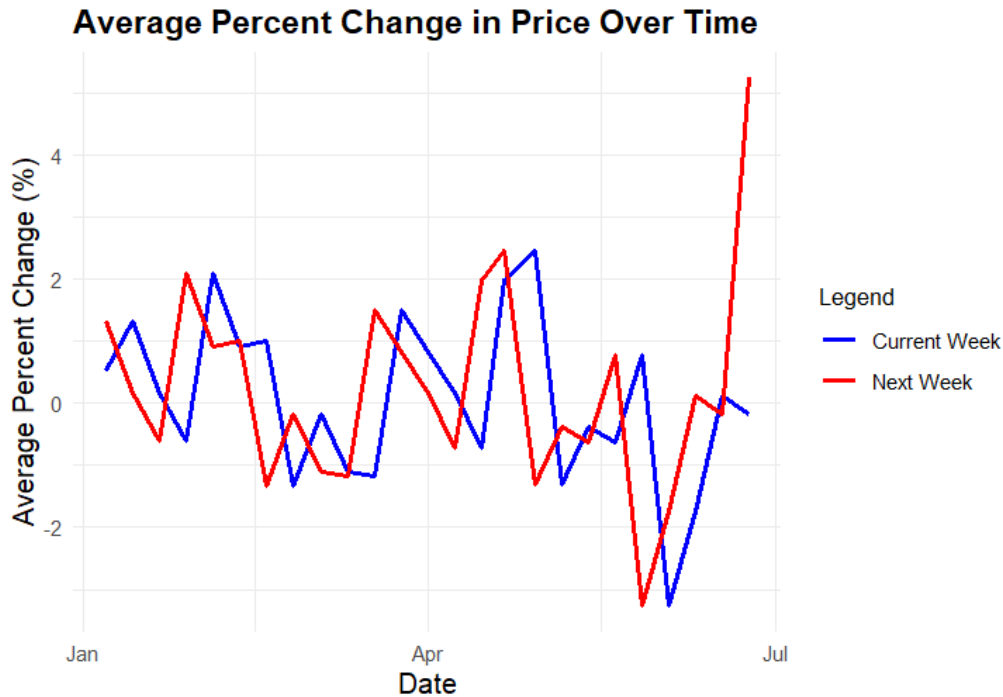


Figure 14: Timeseries graph of average percent change over time.

One might expect that the weeks that have a positive percent change prediction for the following week would happen often. Figure 15 shows the opposite. The boxplot shows that the mean percentage change between the two classes is practically identical. However, one can see that there is a larger spread of values (in both positive and negative directions) for the “1” class. This suggests that there is risk involved in investing via the strategy in this dataset as we can see there is opportunity for both high and low percentage change in price.

Another relationship to be explored is the relationship between percent change in volume and percent change in price, colored by price increase and markers sized by volume (see Figure 16). From visual inspection, there is a saddle point such that high values for percent change in volume and low values in percent change in price have high volume as does low values for percent change in volume and high values in percent change in price. In addition, there appears to be no relationship between the target variable and the other variables in the plot. This suggests that they are not good predictors for the target variable, but may be if interacting with other variables. Again in Figure 17 we see that there is no relationship between the target variable and either of days to next dividend, percent change in price, or their interaction.

To summarise, none of the variables examined appear to be good predictors of the target variable. This is as expected as stock prices are chaotic and require a complex understanding of how many variables interact together.



Figure 15: Boxplot showing how percentage change price varies between the two classes (price increase or no price increase).

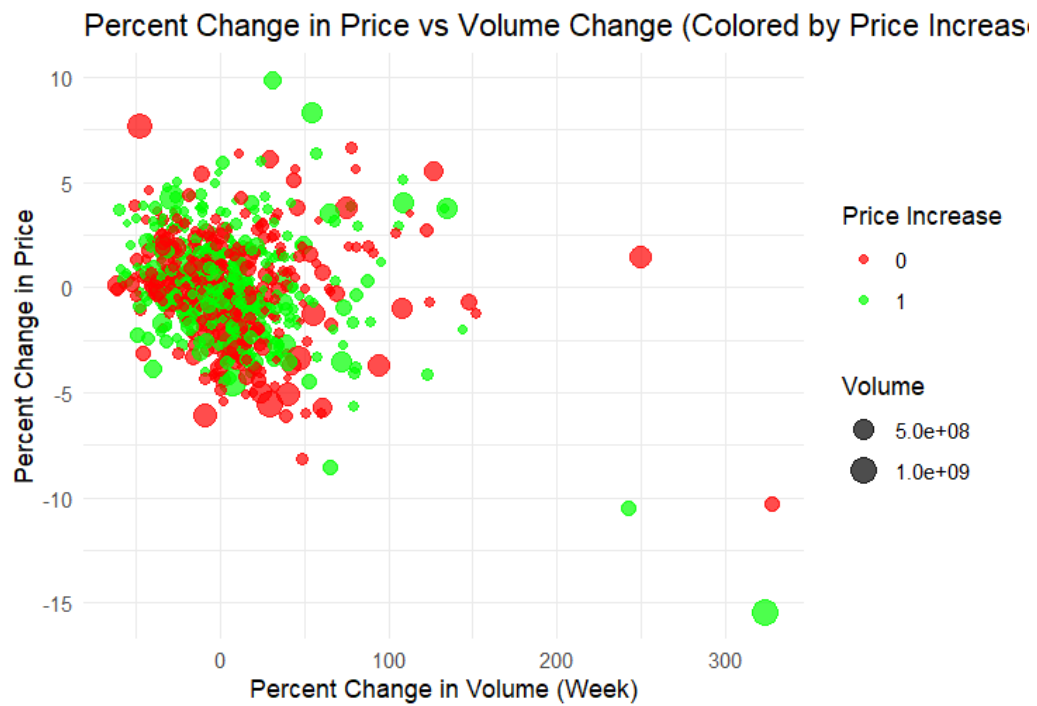


Figure 16: Bubble plot of percent change in price against volume change. Target variable (price increase) is coloured in red and green.

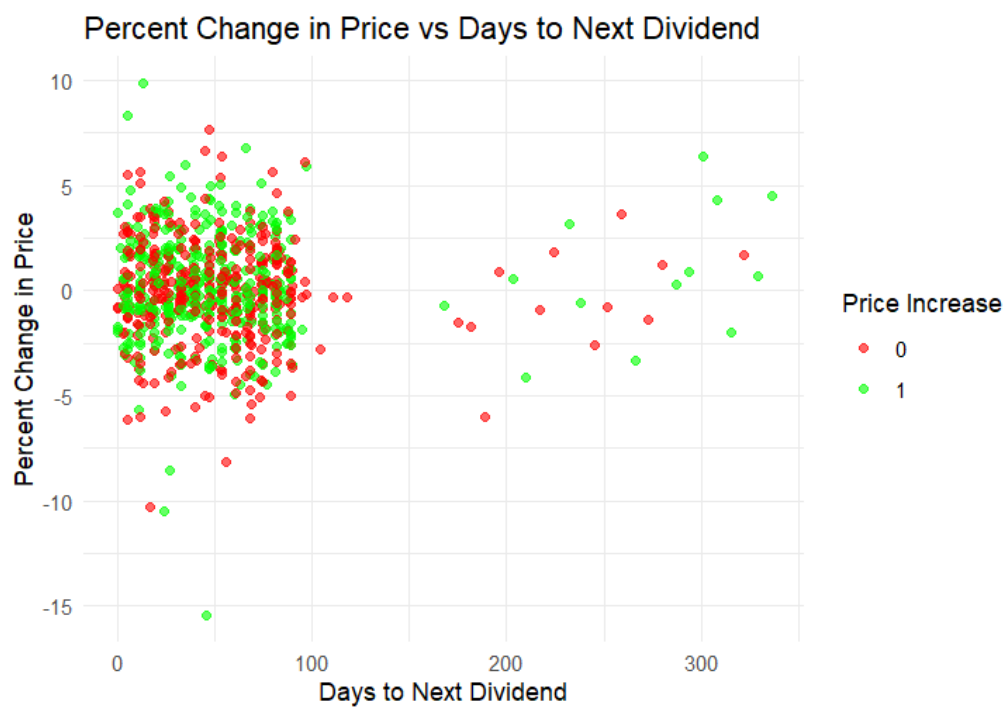


Figure 17: Scatter plot of percent change in price against days to next dividend.

A Appendix: Code Listing

Listing 1: Glass Identification

```
# Loop over all pairs of oxides (including symmetric pairs and diagonal)
for (i in 1:length(num_cols)) {
  for (j in 1:length(num_cols)) {
    if (i != j) { # Exclude diagonal pairs
      # Fit a multiple regression model for RI ~ Oxide1 + Oxide2
      model <- aov(RI ~ ., data = glass[, c("RI", num_cols[i], num_cols[j])])

      # Extract R2
      r2 <- summary(lm(model))$r.squared

      # Store the results
      results_RI <- rbind(
        results_RI,
        data.frame(Oxide1 = num_cols[i], Oxide2 = num_cols[j], R2 = r2)
      )
    }
  }
}
```

Listing 2: El Nino

```
# Correlation Between Air and Sea Surface Temperatures
correlation_value <- cor(el_nino$air.temp., el_nino$s.s.temp., use = "complete.obs")

# Perform Pearson correlation test
cor_test_result <- cor.test(el_nino$air.temp., el_nino$s.s.temp., use = "complete.obs")

# Print the result
print(cor_test_result)

# Scatter Plot with Regression Line and Correlation Value
ggplot(el_nino, aes(x = air.temp., y = s.s.temp.)) +
  geom_point(alpha = 0.5, color = "darkblue") + # Scatter points
  geom_smooth(method = "lm", color = "red", se = FALSE) + # Regression line
  annotate("text", x = 25, y = 22, label = paste("r=", round(correlation_value, 2)),
    size = 5, color = "black") + # Overlay correlation value
  labs(
    title = "Relationship Between Air and Sea Surface Temperatures",
    x = "Air Temperature ( C )",
    y = "Sea Surface Temperature ( C )"
  ) +
  theme_minimal()

# Heatmaps for temperature variations
library(viridis)

# Fixing 'wrap around' in longitude
el_nino$longitude[el_nino$longitude < 0] = el_nino$longitude[el_nino$longitude < 0] + 360

# Air temperature variation by position
ggplot(el_nino, aes(x = longitude, y = latitude, color = air.temp.)) +
  geom_point(size = 2, alpha = 0.7) +
  scale_color_viridis(name = "Air Temp. ( C )") +
  labs(
    title = "Air Temperature by Position",
    x = "Longitude",
    y = "Latitude"
  ) +
  theme_minimal()

# Sea surface temperature variation by position
ggplot(el_nino, aes(x = longitude, y = latitude, color = 's.s.temp.')) +
  geom_point(size = 2, alpha = 0.7) +
  scale_color_viridis(name = "Sea Surface Temp. ( C )") +
  labs(
    title = "Sea Surface Temperature by Position",
    x = "Longitude",
    y = "Latitude"
  ) +
  theme_minimal()

# ANOVA for air temperature by latitude
aov_air_temp <- aov(air.temp. ~ latitude + longitude, data = el_nino)
summary(lm(aov_air_temp))

# ANOVA for sea surface temperature by latitude and longitude
aov_sst <- aov(s.s.temp. ~ latitude + longitude, data = el_nino)
summary(lm(aov_sst))
```

Listing 3: Adult

```
# Scatter plot: Hours per week vs Capital Gain by Income
ggplot(adult_df, aes(x = hours.per.week, y = capital.gain, color = class)) +
  geom_point(alpha = 0.7) +
  labs(title = "Hours-per-Week-vs-Capital-Gain-by-Income", x = "Hours-per-Week",
    y = "Capital-Gain") +
  theme_minimal()

# Income distribution by education
ggplot(adult_df, aes(x = education, fill = class)) +
  geom_bar(position = "fill") +
  labs(title = "Income-Distribution-by-Education-Level", x = "Education-Level",
    y = "Proportion") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

# boxplot for Hours Worked per Week by Education and Income
ggplot(adult_df, aes(x = education, y = hours.per.week, fill = class)) +
  geom_boxplot(outlier.size = 0.5, alpha = 0.7) +
  coord_flip() + # Flip for better readability
  scale_fill_manual(values = c("<=50K" = "blue", ">50K" = "green"), name = "Income") +
  labs(
    title = "Hours-Worked-per-Week-by-Education-and-Income",
    x = "Education-Level",
    y = "Hours-Worked-per-Week"
  ) +
  theme_minimal() +
  theme(axis.text.y = element_text(size = 10))
```

Listing 4: Dow Jones Index

```
dow_jones_index = read.table(file="dow+jones+index/dow_jones_index.data", header=TRUE, sep="
                             col.names=c("quarter", "stock", "date", "open", "high", "low",
                             "volume", "percent_change_price",
                             "percent_change_volume_over_last_week",
                             "previous_weeks_volume", "next_weeks_open",
                             "next_weeks_close", "percent_change_next_weeks_price",
                             "days_to_next_dividend", "percent_return_next_dividend")

dow_jones_index$stock = as.factor(dow_jones_index$stock)
dow_jones_index$date = as.Date(dow_jones_index$date, format="%m/%d/%Y")

# Remove $ sign and convert to numeric in relevant columns
for (col in c(4:7, 12, 13)) {
  dow_jones_index[[col]] = as.numeric(substring(dow_jones_index[[col]], 2))
}

# Scatter plot of percent_change_price vs days_to_next_dividend
ggplot(dow_jones_index, aes(x = days_to_next_dividend, y = percent_change_price, color = factor(percent_change_price > 0))) +
  geom_point(alpha = 0.6) +
  scale_color_manual(values = c("0" = "red", "1" = "green"), name = "Price-Increase") +
  labs(
    title = "Percent-Change-in-Price-vs-Days-to-Next-Dividend",
    x = "Days-to-Next-Dividend",
    y = "Percent-Change-in-Price"
  ) +
  theme_minimal()

# Scatter plot with color-coded price increase
ggplot(dow_jones_index, aes(x = percent_change_volume_over_last_week, y = percent_change_price, color = factor(percent_change_price > 0))) +
  geom_point(alpha = 0.7) +
  scale_color_manual(values = c("0" = "red", "1" = "green"), name = "Price-Increase") +
  labs(
    title = "Percent-Change-in-Price-vs-Volume-Change-(Colored-by-Price-Increase)",
    x = "Percent-Change-in-Volume-(Week)",
    y = "Percent-Change-in-Price",
    size = "Volume"
  ) +
  theme_minimal() +
  theme(legend.position = "right")
```

References

- B. German (1987). *Glass Identification*. DOI: [10.24432/C5WW2P](https://doi.org/10.24432/C5WW2P). URL: <https://archive.ics.uci.edu/dataset/42> Accessed 01/17/2025.
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